

PS C:\Tools\volatility3-win-exes-2.27.0 > .\vol.exe -f

C:\Users\user_lab\Desktop\temp_extract_dir\CYBERDEF-567078-20230213-171333.raw
windows.pslist

Volatility 3 Framework 2.27.0

Progress: 100.00 PDB scanning finished

PID	PPID	ImageFileName	Offset(V)	Threads	Handles	SessionId	Wow64
CreateTime	ExitTime	File output					

4	0	System	0x89c037f8	55	245	N/A	False	N/A	N/A	Disabled
368	4	smss.exe	0x89965020	3	19	N/A	False	2023-02-14		
04:54:15.000000	UTC	N/A	Disabled							
592	368	csrss.exe	0x89a98da0	11	321	0	False	2023-02-14		
04:54:15.000000	UTC	N/A	Disabled							
616	368	winlogon.exe	0x89a88da0	18	508	0	False	2023-02-14		
04:54:15.000000	UTC	N/A	Disabled							
660	616	services.exe	0x89938998	15	240	0	False	2023-02-14		
04:54:15.000000	UTC	N/A	Disabled							
672	616	lsass.exe	0x89aa0020	21	335	0	False	2023-02-14		
04:54:15.000000	UTC	N/A	Disabled							
832	660	VBoxService.exe	0x89aaa3d8	9	115	0	False	2023-02-14		
04:54:15.000000	UTC	N/A	Disabled							
880	660	svchost.exe	0x89aab590	21	295	0	False	2023-02-13		
17:54:16.000000	UTC	N/A	Disabled							
968	660	svchost.exe	0x89a9f6f8	10	244	0	False	2023-02-13		
17:54:17.000000	UTC	N/A	Disabled							
1060	660	svchost.exe	0x89730da0	51	1072	0	False	2023-02-13		
17:54:17.000000	UTC	N/A	Disabled							
1108	660	svchost.exe	0x897289a8	5	78	0	False	2023-02-13		
17:54:17.000000	UTC	N/A	Disabled							
1156	660	svchost.exe	0x899adda0	13	192	0	False	2023-02-13		
17:54:17.000000	UTC	N/A	Disabled							
1484	1440	explorer.exe	0x89733938	14	489	0	False	2023-02-13		
17:54:18.000000	UTC	N/A	Disabled							
1608	660	spoolsv.exe	0x897075d0	10	106	0	False	2023-02-13		
17:54:18.000000	UTC	N/A	Disabled							
480	1060	wscntfy.exe	0x89694388	1	28	0	False	2023-02-13		
17:54:30.000000	UTC	N/A	Disabled							
540	660	alg.exe	0x8969d2a0	5	102	0	False	2023-02-13	17:54:30.000000	
UTC	N/A	Disabled								
376	1484	VBoxTray.exe	0x89982da0	13	125	0	False	2023-02-13		
17:54:30.000000	UTC	N/A	Disabled							
636	1484	msmsgs.exe	0x8994a020	2	157	0	False	2023-02-13		
17:54:30.000000	UTC	N/A	Disabled							
1880	1484	taskmgr.exe	0x89a0b2f0	0	-	0	False	2023-02-13		
18:25:15.000000	UTC	2023-02-13	18:26:21.000000	UTC	Disabled					
964	1484	rootkit.exe	0x899dd740	0	-	0	False	2023-02-13		
18:25:26.000000	UTC	2023-02-13	18:25:26.000000	UTC	Disabled					

```
1960 964 cmd.exe 0x89a18da0 0 - 0 False 2023-02-13
18:25:26.000000 UTC 2023-02-13 18:25:26.000000 UTC Disabled
528 1484 notepad.exe 0x896c5020 0 - 0 False 2023-02-13
18:26:55.000000 UTC 2023-02-13 18:27:46.000000 UTC Disabled
1432 1484 notepad.exe 0x89a0d180 0 - 0 False 2023-02-13
18:28:25.000000 UTC 2023-02-13 18:28:40.000000 UTC Disabled
1444 1484 notepad.exe 0x899e6da0 0 - 0 False 2023-02-13
18:28:42.000000 UTC 2023-02-13 18:28:47.000000 UTC Disabled
276 1484 Dumplt.exe 0x89a0fda0 1 25 0 False 2023-02-13
18:29:08.000000 UTC N/A Disabled
FLARE-VM 04/11/2026 11:51:18
```

```
PS C:\Tools\volatility3-win-exes-2.27.0 > .\vol.exe -f
C:\Users\user_lab\Desktop\temp_extract_dir\CYBERDEF-567078-20230213-171333.raw
windows.info
Volatility 3 Framework 2.27.0
Progress: 100.00 PDB scanning finished
Variable Value

Kernel Base 0x804d7000
DTB 0x39000
Symbols
file:///C:/Tools/volatility3-win-exes-2.27.0/symbols/windows/ntoskrnl.pdb/47A5AC97343A4A7
ABF14EFD9E9933772-2.json.xz
Is64Bit False
IsPAE False
layer_name 0 WindowsIntel
memory_layer 1 FileLayer
KdDebuggerDataBlock 0x8054cde0
NTBuildLab 2600.xpsp.080413-2111
CSDVersion 3
KdVersionBlock 0x8054cdb8
Major/Minor 15.2600
MachineType 332
KeNumberProcessors 1
SystemTime 2023-02-13 18:29:11+00:00
NtSystemRoot C:\WINDOWS
NtProductType NtProductWinNt
NtMajorVersion 5
NtMinorVersion 1
PE MajorOperatingSystemVersion 5
PE MinorOperatingSystemVersion 1
PE Machine 332
PE TimeDateStamp Sun Apr 13 19:27:39 2008
FLARE-VM 04/11/2026 11:53:33
```

```
PS C:\Tools\volatility3-win-exes-2.27.0 > .\vol.exe -f
C:\Users\user_lab\Desktop\temp_extract_dir\CYBERDEF-567078-20230213-171333.raw
windows.info
Volatility 3 Framework 2.27.0
Progress: 100.00          PDB scanning finished
Variable      Value

Kernel Base    0x804d7000
DTB            0x39000
Symbols
file:///C:/Tools/volatility3-win-exes-2.27.0/symbols/windows/ntoskrnl.pdb/47A5AC97343A4A7
ABF14EFD9E9933772-2.json.xz
Is64Bit False
IsPAE False
layer_name     0 WindowsIntel
memory_layer   1 FileLayer
KdDebuggerDataBlock 0x8054cde0
NTBuildLab     2600.xpsp.080413-2111
CSDVersion     3
KdVersionBlock 0x8054cdb8
Major/Minor    15.2600
MachineType    332
KeNumberProcessors 1
SystemTime     2023-02-13 18:29:11+00:00
NtSystemRoot   C:\WINDOWS
NtProductType  NtProductWinNt
NtMajorVersion 5
NtMinorVersion 1
PE MajorOperatingSystemVersion 5
PE MinorOperatingSystemVersion 1
PE Machine     332
PE TimeDateStamp    Sun Apr 13 19:27:39 2008
FLARE-VM 04/11/2026 11:58:41
PS C:\Tools\volatility3-win-exes-2.27.0 > .\vol.exe -f
C:\Users\user_lab\Desktop\temp_extract_dir\CYBERDEF-567078-20230213-171333.raw
windows.pslist | find /c /v ""
FIND: Parameter format not correctB scanning finished
FLARE-VM 04/11/2026 12:03:31
PS C:\Tools\volatility3-win-exes-2.27.0 > .\vol.exe -f
C:\Users\user_lab\Desktop\temp_extract_dir\CYBERDEF-567078-20230213-171333.raw
windows.malfind
```

Volatility 3 Framework 2.27.0

volatility3\framework\deprecation.py:28: FutureWarning: This API (volatility3.plugins.windows.malware.malfind.Malfind.run) will be removed in the first release after 2026-06-07. This plugin has been renamed, please call volatility3.plugins.windows.malware.malfind.Malfind rather than volatility3.plugins.windows.malfind.Malfind.

PID Process Start VPN End VPN Tag Protection CommitCharge
PrivateMemory File output Notes Hexdump Disasm
volatility3\framework\deprecation.py:105: FutureWarning: This plugin (volatility3.plugins.windows.malfind.Malfind) has been renamed and will be removed in the first release after 2026-06-07. Please ensure all method calls to this plugin are replaced with calls to volatility3.plugins.windows.malware.malfind.Malfind

```
592  csrss.exe  0x7f6f0000  0x7f7effff  Vad  PAGE_EXECUTE_READWRITE  0
0    Disabled  N/A
c8 00 00 00 84 01 00 00 ff ee ff ee 08 70 00 00 .....p..
08 00 00 00 00 fe 00 00 00 00 10 00 00 20 00 00 ..... ..
00 02 00 00 00 20 00 00 8d 01 00 00 ff ef fd 7f .....
03 00 08 06 00 00 00 00 00 00 00 00 00 00 00 00 .....
0x7f6f0000:  enter  0, 0
0x7f6f0004:  test   byte ptr [ecx], al
0x7f6f0006:  add    byte ptr [eax], al
616  winlogon.exe 0x580000  0x59ffff  Vad  PAGE_EXECUTE_READ  0
0    Disabled  N/A
c1 00 00 00 00 01 00 00 ff ee ff ee 09 00 00 00 .....
09 00 00 00 00 fe 00 00 00 00 10 00 00 20 00 00 ..... ..
00 02 00 00 00 20 00 00 47 08 00 00 ff ef fd 7f ..... ..G.....
00 00 08 06 00 00 00 00 00 00 00 00 00 00 00 00 .....
0x580000:  rol    dword ptr [eax], 0
0x580003:  add    byte ptr [eax], al
0x580005:  add    dword ptr [eax], eax
0x580007:  add    bh, bh
0x580009:  out    dx, al
616  winlogon.exe 0x4a5e0000 0x4a5e3fff  VadS
PAGE_EXECUTE_READWRITE 4  1  Disabled  N/A
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00 00 00 00 25 00 25 00 01 00 00 00 00 00 00 00 ....%.%.....
0x4a5e0000:  add    byte ptr [eax], al
0x4a5e0002:  add    byte ptr [eax], al
0x4a5e0004:  add    byte ptr [eax], al
0x4a5e0006:  add    byte ptr [eax], al
0x4a5e0008:  add    byte ptr [eax], al
0x4a5e000a:  add    byte ptr [eax], al
0x4a5e000c:  add    byte ptr [eax], al
0x4a5e000e:  add    byte ptr [eax], al
```

```

0x4a5e0010: add byte ptr [eax], al
0x4a5e0012: add byte ptr [eax], al
0x4a5e0014: add byte ptr [eax], al
0x4a5e0016: add byte ptr [eax], al
0x4a5e0018: add byte ptr [eax], al
0x4a5e001a: add byte ptr [eax], al
0x4a5e001c: add byte ptr [eax], al
0x4a5e001e: add byte ptr [eax], al
0x4a5e0020: add byte ptr [eax], al
0x4a5e0022: add byte ptr [eax], al
0x4a5e0024: add byte ptr [eax], al
0x4a5e0026: add byte ptr [eax], al
0x4a5e0028: add byte ptr [eax], al
0x4a5e002a: add byte ptr [eax], al
0x4a5e002c: add byte ptr [eax], al
0x4a5e002e: add byte ptr [eax], al
0x4a5e0030: add byte ptr [eax], al
0x4a5e0032: add byte ptr [eax], al
0x4a5e0034: and eax, 0x1002500
0x4a5e0039: add byte ptr [eax], al
0x4a5e003b: add byte ptr [eax], al
0x4a5e003d: add byte ptr [eax], al

```

```

616 winlogon.exe 0x2afc0000 0x2afc3fff VadS PAGE_EXECUTE_READWRITE
4 1 Disabled N/A

```

```

00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00 00 00 00 23 00 23 00 01 00 00 00 00 00 00 00 ....##.....

```

```

0x2afc0000: add byte ptr [eax], al
0x2afc0002: add byte ptr [eax], al
0x2afc0004: add byte ptr [eax], al
0x2afc0006: add byte ptr [eax], al
0x2afc0008: add byte ptr [eax], al
0x2afc000a: add byte ptr [eax], al
0x2afc000c: add byte ptr [eax], al
0x2afc000e: add byte ptr [eax], al
0x2afc0010: add byte ptr [eax], al
0x2afc0012: add byte ptr [eax], al
0x2afc0014: add byte ptr [eax], al
0x2afc0016: add byte ptr [eax], al
0x2afc0018: add byte ptr [eax], al
0x2afc001a: add byte ptr [eax], al
0x2afc001c: add byte ptr [eax], al
0x2afc001e: add byte ptr [eax], al
0x2afc0020: add byte ptr [eax], al
0x2afc0022: add byte ptr [eax], al
0x2afc0024: add byte ptr [eax], al
0x2afc0026: add byte ptr [eax], al

```

```

0x2afc0028:  add    byte ptr [eax], al
0x2afc002a:  add    byte ptr [eax], al
0x2afc002c:  add    byte ptr [eax], al
0x2afc002e:  add    byte ptr [eax], al
0x2afc0030:  add    byte ptr [eax], al
0x2afc0032:  add    byte ptr [eax], al
0x2afc0034:  and     eax, dword ptr [eax]
0x2afc0036:  and     eax, dword ptr [eax]
0x2afc0038:  add     dword ptr [eax], eax
0x2afc003a:  add    byte ptr [eax], al
0x2afc003c:  add    byte ptr [eax], al
0x2afc003e:  add    byte ptr [eax], al
616  winlogon.exe  0xe880000  0xe883fff  VadS  PAGE_EXECUTE_READWRITE
4    1    Disabled  N/A
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00 00 00 00 25 00 25 00 01 00 00 00 00 00 00 00 ....%.%.
0xe880000:  add    byte ptr [eax], al
0xe880002:  add    byte ptr [eax], al
0xe880004:  add    byte ptr [eax], al
0xe880006:  add    byte ptr [eax], al
0xe880008:  add    byte ptr [eax], al
0xe88000a:  add    byte ptr [eax], al
0xe88000c:  add    byte ptr [eax], al
0xe88000e:  add    byte ptr [eax], al
0xe880010:  add    byte ptr [eax], al
0xe880012:  add    byte ptr [eax], al
0xe880014:  add    byte ptr [eax], al
0xe880016:  add    byte ptr [eax], al
0xe880018:  add    byte ptr [eax], al
0xe88001a:  add    byte ptr [eax], al
0xe88001c:  add    byte ptr [eax], al
0xe88001e:  add    byte ptr [eax], al
0xe880020:  add    byte ptr [eax], al
0xe880022:  add    byte ptr [eax], al
0xe880024:  add    byte ptr [eax], al
0xe880026:  add    byte ptr [eax], al
0xe880028:  add    byte ptr [eax], al
0xe88002a:  add    byte ptr [eax], al
0xe88002c:  add    byte ptr [eax], al
0xe88002e:  add    byte ptr [eax], al
0xe880030:  add    byte ptr [eax], al
0xe880032:  add    byte ptr [eax], al
0xe880034:  and     eax, 0x1002500
0xe880039:  add    byte ptr [eax], al
0xe88003b:  add    byte ptr [eax], al
0xe88003d:  add    byte ptr [eax], al

```

```

616  winlogon.exe  0x16be0000  0x16be3fff  VadS
PAGE_EXECUTE_READWRITE 4    1    Disabled    N/A
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00 00 00 00 28 00 28 00 01 00 00 00 00 00 00 00 ....(.(.....
0x16be0000:  add    byte ptr [eax], al
0x16be0002:  add    byte ptr [eax], al
0x16be0004:  add    byte ptr [eax], al
0x16be0006:  add    byte ptr [eax], al
0x16be0008:  add    byte ptr [eax], al
0x16be000a:  add    byte ptr [eax], al
0x16be000c:  add    byte ptr [eax], al
0x16be000e:  add    byte ptr [eax], al
0x16be0010:  add    byte ptr [eax], al
0x16be0012:  add    byte ptr [eax], al
0x16be0014:  add    byte ptr [eax], al
0x16be0016:  add    byte ptr [eax], al
0x16be0018:  add    byte ptr [eax], al
0x16be001a:  add    byte ptr [eax], al
0x16be001c:  add    byte ptr [eax], al
0x16be001e:  add    byte ptr [eax], al
0x16be0020:  add    byte ptr [eax], al
0x16be0022:  add    byte ptr [eax], al
0x16be0024:  add    byte ptr [eax], al
0x16be0026:  add    byte ptr [eax], al
0x16be0028:  add    byte ptr [eax], al
0x16be002a:  add    byte ptr [eax], al
0x16be002c:  add    byte ptr [eax], al
0x16be002e:  add    byte ptr [eax], al
0x16be0030:  add    byte ptr [eax], al
0x16be0032:  add    byte ptr [eax], al
0x16be0034:  sub     byte ptr [eax], al
0x16be0036:  sub     byte ptr [eax], al
0x16be0038:  add     dword ptr [eax], eax
0x16be003a:  add     byte ptr [eax], al
0x16be003c:  add     byte ptr [eax], al
0x16be003e:  add     byte ptr [eax], al
616  winlogon.exe  0x2b010000  0x2b013fff  VadS
PAGE_EXECUTE_READWRITE 4    1    Disabled    N/A
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00 00 00 00 2c 00 2c 00 01 00 00 00 00 00 00 00 ....,.,.....
0x2b010000:  add     byte ptr [eax], al
0x2b010002:  add     byte ptr [eax], al
0x2b010004:  add     byte ptr [eax], al
0x2b010006:  add     byte ptr [eax], al

```

```

0x2b010008: add byte ptr [eax], al
0x2b01000a: add byte ptr [eax], al
0x2b01000c: add byte ptr [eax], al
0x2b01000e: add byte ptr [eax], al
0x2b010010: add byte ptr [eax], al
0x2b010012: add byte ptr [eax], al
0x2b010014: add byte ptr [eax], al
0x2b010016: add byte ptr [eax], al
0x2b010018: add byte ptr [eax], al
0x2b01001a: add byte ptr [eax], al
0x2b01001c: add byte ptr [eax], al
0x2b01001e: add byte ptr [eax], al
0x2b010020: add byte ptr [eax], al
0x2b010022: add byte ptr [eax], al
0x2b010024: add byte ptr [eax], al
0x2b010026: add byte ptr [eax], al
0x2b010028: add byte ptr [eax], al
0x2b01002a: add byte ptr [eax], al
0x2b01002c: add byte ptr [eax], al
0x2b01002e: add byte ptr [eax], al
0x2b010030: add byte ptr [eax], al
0x2b010032: add byte ptr [eax], al
0x2b010034: sub al, 0
0x2b010036: sub al, 0
0x2b010038: add dword ptr [eax], eax
0x2b01003a: add byte ptr [eax], al
0x2b01003c: add byte ptr [eax], al
0x2b01003e: add byte ptr [eax], al
616 winlogon.exe 0x3ad10000 0x3ad13fff VadS
PAGE_EXECUTE_READWRITE 4 1 Disabled N/A
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00 00 00 00 25 00 25 00 01 00 00 00 00 00 00 00 ....%.%.....
0x3ad10000: add byte ptr [eax], al
0x3ad10002: add byte ptr [eax], al
0x3ad10004: add byte ptr [eax], al
0x3ad10006: add byte ptr [eax], al
0x3ad10008: add byte ptr [eax], al
0x3ad1000a: add byte ptr [eax], al
0x3ad1000c: add byte ptr [eax], al
0x3ad1000e: add byte ptr [eax], al
0x3ad10010: add byte ptr [eax], al
0x3ad10012: add byte ptr [eax], al
0x3ad10014: add byte ptr [eax], al
0x3ad10016: add byte ptr [eax], al
0x3ad10018: add byte ptr [eax], al
0x3ad1001a: add byte ptr [eax], al

```



```

0x3ad1001c:  add    byte ptr [eax], al
0x3ad1001e:  add    byte ptr [eax], al
0x3ad10020:  add    byte ptr [eax], al
0x3ad10022:  add    byte ptr [eax], al
0x3ad10024:  add    byte ptr [eax], al
0x3ad10026:  add    byte ptr [eax], al
0x3ad10028:  add    byte ptr [eax], al
0x3ad1002a:  add    byte ptr [eax], al
0x3ad1002c:  add    byte ptr [eax], al
0x3ad1002e:  add    byte ptr [eax], al
0x3ad10030:  add    byte ptr [eax], al
0x3ad10032:  add    byte ptr [eax], al
0x3ad10034:  and     eax, 0x1002500
0x3ad10039:  add    byte ptr [eax], al
0x3ad1003b:  add    byte ptr [eax], al
0x3ad1003d:  add    byte ptr [eax], al
616  winlogon.exe  0x3f750000  0x3f753fff  VadS  PAGE_EXECUTE_READWRITE
4    1    Disabled  N/A
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00 00 00 00 27 00 27 00 01 00 00 00 00 00 00 00 ....'!'.....
0x3f750000:  add    byte ptr [eax], al
0x3f750002:  add    byte ptr [eax], al
0x3f750004:  add    byte ptr [eax], al
0x3f750006:  add    byte ptr [eax], al
0x3f750008:  add    byte ptr [eax], al
0x3f75000a:  add    byte ptr [eax], al
0x3f75000c:  add    byte ptr [eax], al
0x3f75000e:  add    byte ptr [eax], al
0x3f750010:  add    byte ptr [eax], al
0x3f750012:  add    byte ptr [eax], al
0x3f750014:  add    byte ptr [eax], al
0x3f750016:  add    byte ptr [eax], al
0x3f750018:  add    byte ptr [eax], al
0x3f75001a:  add    byte ptr [eax], al
0x3f75001c:  add    byte ptr [eax], al
0x3f75001e:  add    byte ptr [eax], al
0x3f750020:  add    byte ptr [eax], al
0x3f750022:  add    byte ptr [eax], al
0x3f750024:  add    byte ptr [eax], al
0x3f750026:  add    byte ptr [eax], al
0x3f750028:  add    byte ptr [eax], al
0x3f75002a:  add    byte ptr [eax], al
0x3f75002c:  add    byte ptr [eax], al
0x3f75002e:  add    byte ptr [eax], al
0x3f750030:  add    byte ptr [eax], al
0x3f750032:  add    byte ptr [eax], al

```

```

0x3f750034:  daa
0x3f750035:  add  byte ptr [edi], ah
0x3f750037:  add  byte ptr [ecx], al
0x3f750039:  add  byte ptr [eax], al
0x3f75003b:  add  byte ptr [eax], al
0x3f75003d:  add  byte ptr [eax], al
616  winlogon.exe  0x5d080000  0x5d083fff  VadS
PAGE_EXECUTE_READWRITE 4  1  Disabled  N/A
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00 00 00 00 28 00 28 00 01 00 00 00 00 00 00 00 ....(.(.....
0x5d080000:  add  byte ptr [eax], al
0x5d080002:  add  byte ptr [eax], al
0x5d080004:  add  byte ptr [eax], al
0x5d080006:  add  byte ptr [eax], al
0x5d080008:  add  byte ptr [eax], al
0x5d08000a:  add  byte ptr [eax], al
0x5d08000c:  add  byte ptr [eax], al
0x5d08000e:  add  byte ptr [eax], al
0x5d080010:  add  byte ptr [eax], al
0x5d080012:  add  byte ptr [eax], al
0x5d080014:  add  byte ptr [eax], al
0x5d080016:  add  byte ptr [eax], al
0x5d080018:  add  byte ptr [eax], al
0x5d08001a:  add  byte ptr [eax], al
0x5d08001c:  add  byte ptr [eax], al
0x5d08001e:  add  byte ptr [eax], al
0x5d080020:  add  byte ptr [eax], al
0x5d080022:  add  byte ptr [eax], al
0x5d080024:  add  byte ptr [eax], al
0x5d080026:  add  byte ptr [eax], al
0x5d080028:  add  byte ptr [eax], al
0x5d08002a:  add  byte ptr [eax], al
0x5d08002c:  add  byte ptr [eax], al
0x5d08002e:  add  byte ptr [eax], al
0x5d080030:  add  byte ptr [eax], al
0x5d080032:  add  byte ptr [eax], al
0x5d080034:  sub  byte ptr [eax], al
0x5d080036:  sub  byte ptr [eax], al
0x5d080038:  add  dword ptr [eax], eax
0x5d08003a:  add  byte ptr [eax], al
0x5d08003c:  add  byte ptr [eax], al
0x5d08003e:  add  byte ptr [eax], al
616  winlogon.exe  0x62220000  0x62223fff  VadS
PAGE_EXECUTE_READWRITE 4  1  Disabled  N/A
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....

```

00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 2a 00 2a 00 01 00 00 00 00 00 00 00*.*.....

0x62220000: add byte ptr [eax], al
0x62220002: add byte ptr [eax], al
0x62220004: add byte ptr [eax], al
0x62220006: add byte ptr [eax], al
0x62220008: add byte ptr [eax], al
0x6222000a: add byte ptr [eax], al
0x6222000c: add byte ptr [eax], al
0x6222000e: add byte ptr [eax], al
0x62220010: add byte ptr [eax], al
0x62220012: add byte ptr [eax], al
0x62220014: add byte ptr [eax], al
0x62220016: add byte ptr [eax], al
0x62220018: add byte ptr [eax], al
0x6222001a: add byte ptr [eax], al
0x6222001c: add byte ptr [eax], al
0x6222001e: add byte ptr [eax], al
0x62220020: add byte ptr [eax], al
0x62220022: add byte ptr [eax], al
0x62220024: add byte ptr [eax], al
0x62220026: add byte ptr [eax], al
0x62220028: add byte ptr [eax], al
0x6222002a: add byte ptr [eax], al
0x6222002c: add byte ptr [eax], al
0x6222002e: add byte ptr [eax], al
0x62220030: add byte ptr [eax], al
0x62220032: add byte ptr [eax], al
0x62220034: sub al, byte ptr [eax]
0x62220036: sub al, byte ptr [eax]
0x62220038: add dword ptr [eax], eax
0x6222003a: add byte ptr [eax], al
0x6222003c: add byte ptr [eax], al
0x6222003e: add byte ptr [eax], al

616 winlogon.exe 0x7f6f0000 0x7f7effff Vad PAGE_EXECUTE_READ 0

0 Disabled N/A

c8 00 00 00 84 01 00 00 ff ee ff ee 08 70 00 00p..
08 00 00 00 00 fe 00 00 00 00 10 00 00 20 00 00
00 02 00 00 00 20 00 00 8d 01 00 00 ff ef fd 7f
03 00 08 06 00 00 00 00 00 00 00 00 00 00 00 00

0x7f6f0000: enter 0, 0
0x7f6f0004: test byte ptr [ecx], al
0x7f6f0006: add byte ptr [eax], al

880 svchost.exe **0x980000** 0x988fff VadS PAGE_EXECUTE_READWRITE

9 1 Disabled MZ header

4d 5a 90 00 03 00 00 00 04 00 00 00 ff ff 00 00 MZ.....

b8 00 00 00 00 00 00 00 40 00 00 00 00 00 00 00@.....
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

```

00 00 00 00 00 00 00 00 00 00 00 00 f8 00 00 00 .....
0x980000:    dec    ebp
0x980001:    pop    edx
0x980002:    nop
0x980003:    add    byte ptr [ebx], al
0x980005:    add    byte ptr [eax], al
0x980007:    add    byte ptr [eax + eax], al
0x98000a:    add    byte ptr [eax], al
636  msmsgs.exe  0x490000    0x78ffff    Vad  PAGE_EXECUTE_READ  0
0    Disabled    N/A
c1 00 00 00 00 01 00 00 ff ee ff ee 09 00 00 00 .....
09 00 00 00 00 fe 00 00 00 00 10 00 00 20 00 00 ..... ..
00 02 00 00 00 20 00 00 1b 1f 00 00 ff ef fd 7f .....
00 00 08 06 00 00 00 00 00 00 00 00 00 00 00 00 .....
0x490000:    rol    dword ptr [eax], 0
0x490003:    add    byte ptr [eax], al
0x490005:    add    dword ptr [eax], eax
0x490007:    add    bh, bh
0x490009:    out    dx, al
FLARE-VM 04/11/2026 12:09:39

```

PS C:\Tools\volatility3-win-exes-2.27.0 > .\vol.exe -f
C:\Users\user_lab\Desktop\temp_extract_dir\CYBERDEF-567078-20230213-171333.raw
windows.handles.Handles --pid 880

Volatility 3 Framework 2.27.0

Progress: 100.00 PDB scanning finished

PID	Process	Offset	HandleValue	Type	GrantedAccess	Name
-----	---------	--------	-------------	------	---------------	------

880	svchost.exe	0xe10096d0	0x4	KeyedEvent	0xf0003	CritSecOutOfMemoryEvent
880	svchost.exe	0xe15bfb10	0x8	Directory	0x3	KnownDlls
880	svchost.exe	0x89a28890	0xc	File	0x100020	\Device\HarddiskVolume1\WINDOWS\system32
880	svchost.exe	0x899372e0	0x10	Semaphore	0x100003	-
880	svchost.exe	0xe15cf880	0x14	Directory	0xf000f	Windows
880	svchost.exe	0xe16e2f68	0x18	Port	0x21f0001	-
880	svchost.exe	0x89937318	0x1c	Semaphore	0x100003	-
880	svchost.exe	0xe1439c68	0x20	Directory	0x2000f	BaseNamedObjects
880	svchost.exe	0x8996a148	0x24	Mutant	0x1f0001	SHIMLIB_LOG_MUTEX
880	svchost.exe	0xe164c850	0x28	Key	0x20f003f	MACHINE
880	svchost.exe	0x899372b0	0x2c	Event	0x21f0003	-
880	svchost.exe	0x89af33e8	0x30	WindowStation	0xf016e	Service-0x0-3e7\$
880	svchost.exe	0x89b42b18	0x34	Desktop	0xf00cf	Default
880	svchost.exe	0x89af33e8	0x38	WindowStation	0xf016e	Service-0x0-3e7\$

```

880  svchost.exe  0x899d8968  0x3c  Event  0x1f0003  -
880  svchost.exe  0x89a93cd0  0x40  Semaphore  0x100003  -
880  svchost.exe  0x89a93c98  0x44  Semaphore  0x100003  -
880  svchost.exe  0xe164eeb8  0x48  Key  0x20019
MACHINE\SOFTWARE\MICROSOFT\WINDOWS NT\CURRENTVERSION\DRIVERS32
880  svchost.exe  0x8997cd40  0x4c  Event  0x1f0003  DINPUTWINMM
880  svchost.exe  0x89a1a6f8  0x50  File  0x100001  \Device\KsecDD
880  svchost.exe  0x89a93c68  0x54  Event  0x1f0003  -
880  svchost.exe  0x899f57a0  0x58  Event  0x1f0003  -
880  svchost.exe  0xe16e4e40  0x5c  Key  0x20019
MACHINE\SOFTWARE\MICROSOFT\WINDOWS NT\CURRENTVERSION\DRIVERS32
880  svchost.exe  0x89af1b70  0x60  Semaphore  0x1f0003
shell.{A48F1A32-A340-11D1-BC6B-00A0C90312E1}
880  svchost.exe  0x89a8e1f0  0x64  Event  0x1f0003  userenv: User Profile
setup event
880  svchost.exe  0x89937358  0x68  File  0x100020
\Device\HarddiskVolume1\WINDOWS\WinSxS\x86_Microsoft.Windows.Common-Controls_6
595b64144ccf1df_6.0.2600.5512_x-ww_35d4ce83
880  svchost.exe  0xe163c1f0  0x6c  Token  0x8  -
880  svchost.exe  0x89a15710  0x70  Event  0x1f0003  -
880  svchost.exe  0x89a156d8  0x74  Semaphore  0x100003  -
880  svchost.exe  0x89a148f0  0x78  Semaphore  0x100003  -
880  svchost.exe  0x89a148b8  0x7c  Semaphore  0x100003  -
880  svchost.exe  0x89a141c0  0x80  Semaphore  0x100003  -
880  svchost.exe  0x89a14188  0x84  Semaphore  0x100003  -
880  svchost.exe  0x89a13450  0x88  Semaphore  0x100003  -
880  svchost.exe  0x89a13418  0x8c  Semaphore  0x100003  -
880  svchost.exe  0x89a11788  0x90  Semaphore  0x100003  -
880  svchost.exe  0x89a11750  0x94  Semaphore  0x100003  -
880  svchost.exe  0x89a0ba80  0x98  Semaphore  0x100003  -
880  svchost.exe  0x89a0ba48  0x9c  Semaphore  0x100003  -
880  svchost.exe  0x899da7a0  0xa0  Semaphore  0x100003  -
880  svchost.exe  0x89a194c8  0xa4  Event  0x1f0003  -
880  svchost.exe  0x89a0ba18  0xa8  Event  0x1f0003  -
880  svchost.exe  0x89a19498  0xac  Event  0x1f0003  -
880  svchost.exe  0x89a19468  0xb0  Event  0x1f0003  -
880  svchost.exe  0x89aab318  0xb4  Thread  0x1f03ff  Tid 884 Pid 880
880  svchost.exe  0x89a18820  0xb8  Event  0x1f0003  -
880  svchost.exe  0x899d0250  0xbc  File  0x12019f
\Device\NamedPipe\nt\NtControlPipe2
880  svchost.exe  0xe1681f68  0xc0  Port  0x1f0001  -
880  svchost.exe  0x8972eda8  0xc4  Thread  0x1f03ff  Tid 1068 Pid 880
880  svchost.exe  0x89953318  0xc8  Event  0x1f0003  -
880  svchost.exe  0x89730ca0  0xcc  Event  0x1f0003  -
880  svchost.exe  0x89953348  0xd0  Event  0x1f0003  -
880  svchost.exe  0x89955020  0xd4  Event  0x1f0003  -
880  svchost.exe  0x899e37a0  0xd8  IoCompletion  0x1f0003  -
880  svchost.exe  0x899d02e0  0xdc  IoCompletion  0x1f0003  -

```

880	svchost.exe	0x899e37a0	0xe0	IoCompletion	0x1f0003	-
880	svchost.exe	0xe16a84f8	0xe4	Key	0x20f003f	
MACHINE\SOFTWARE\CLASSES						
880	svchost.exe	0xe1676020	0xe8	Key	0x20019	
MACHINE\SOFTWARE\CLASSES\CLSID						
880	svchost.exe	0xe1683530	0xec	Key	0x20019	
MACHINE\SOFTWARE\CLASSES\APPID						
880	svchost.exe	0x899550f0	0xf0	Event	0x1f0003	-
880	svchost.exe	0x899550c0	0xf4	Event	0x1f0003	-
880	svchost.exe	0x89aa41b8	0xf8	Event	0x1f0003	-
880	svchost.exe	0x89aa4188	0xfc	Event	0x1f0003	-
880	svchost.exe	0x89a17a50	0x100	File	0x100000	\Device\Dfs
880	svchost.exe	0xe16e8518	0x104	Port	0x1f0001	-
880	svchost.exe	0x89998940	0x108	Event	0x1f0003	-
880	svchost.exe	0xe1693758	0x10c	Port	0x1f0001	-
880	svchost.exe	0xe16e7eb0	0x110	Key	0x20019	
MACHINE\SOFTWARE\MICROSOFT\OLE						
880	svchost.exe	0xe1684b78	0x114	Key	0x10	
MACHINE\SOFTWARE\MICROSOFT\OLE						
880	svchost.exe	0x89999620	0x118	Timer	0x1f0003	-
880	svchost.exe	0x89955060	0x11c	Event	0x21f0003	-
880	svchost.exe	0x8994d988	0x120	Thread	0x1f03ff	Tid 924 Pid 880
880	svchost.exe	0x899995d8	0x124	IoCompletion	0x1f0003	-
880	svchost.exe	0x89a2c2b0	0x128	Timer	0x1f0003	-
880	svchost.exe	0x89949a90	0x12c	Thread	0x1f03ff	Tid 932 Pid 880
880	svchost.exe	0xe16a6b08	0x130	Key	0x20019	
MACHINE\SOFTWARE\POLICIES						
880	svchost.exe	0x89998910	0x134	Event	0x1f0003	-
880	svchost.exe	0xe16907e8	0x138	Key	0x10	
MACHINE\SOFTWARE\POLICIES						
880	svchost.exe	0xe168ef68	0x13c	Port	0x1f0001	actkernel
880	svchost.exe	0x89a1f890	0x140	Event	0x1f0003	-
880	svchost.exe	0x8994e240	0x144	Thread	0x1f03ff	Tid 936 Pid 880
880	svchost.exe	0x89a9f6f8	0x148	Process	0x1f0fff	svchost.exe Pid 968
880	svchost.exe	0xe1744e60	0x14c	Section	0xf0007	RotHintTable
880	svchost.exe	0xe17373d8	0x150	Port	0x1f0001	-
880	svchost.exe	0x8999f3e0	0x154	Event	0x1f0003	-
880	svchost.exe	0x89732cb8	0x158	File	0x12019f	
\Device\NamedPipe\lsarpc						
880	svchost.exe	0x8972ed38	0x15c	Mutant	0x1f0001	
{A3BD3259-3E4F-428a-84C8-F0463A9D3EB5}						
880	svchost.exe	0xe173a3d0	0x160	Section	0xf0007	
{A64C7F33-DA35-459b-96CA-63B51FB0CDB9}						
880	svchost.exe	0x89693888	0x164	Event	0x1f0003	-
880	svchost.exe	0xe16b2d98	0x168	Key	0xf003f	
MACHINE\SOFTWARE\CLASSES						
880	svchost.exe	0xe19821a0	0x16c	Key	0xf003f	
MACHINE\SOFTWARE\CLASSES						

880	svchost.exe	0x899634b8	0x170	Event	0x1f0003	-
880	svchost.exe	0xe1743330	0x174	Key	0xf003f	
MACHINE\SOFTWARE\MICROSOFT\COM3						
880	svchost.exe	0x89963488	0x178	Event	0x1f0003	-
880	svchost.exe	0xe17431b8	0x17c	Key	0x10 USER	
880	svchost.exe	0x89715d78	0x180	Event	0x1f0003	-
880	svchost.exe	0xe187de40	0x184	Key	0xf003f	
MACHINE\SOFTWARE\CLASSES						
880	svchost.exe	0x89715d10	0x188	Event	0x1f0003	-
880	svchost.exe	0xe1703e40	0x18c	Key	0x10 USER	
880	svchost.exe	0x89715ca8	0x190	Event	0x1f0003	-
880	svchost.exe	0xe1a7a870	0x194	Key	0xf003f	
MACHINE\SOFTWARE\MICROSOFT\COM3						
880	svchost.exe	0x899c9fb8	0x198	Event	0x1f0003	-
880	svchost.exe	0xe1941cc8	0x19c	Key	0xf003f	
MACHINE\SOFTWARE\MICROSOFT\COM3						
880	svchost.exe	0x899c9f50	0x1a0	Event	0x1f0003	-
880	svchost.exe	0xe19422a8	0x1a4	Key	0xf003f	
MACHINE\SOFTWARE\CLASSES\CLSID						
880	svchost.exe	0x89967ef8	0x1a8	Event	0x1f0003	-
880	svchost.exe	0xe1a94458	0x1ac	Key	0xf003f	
MACHINE\SOFTWARE\CLASSES						
880	svchost.exe	0x89967e90	0x1b0	Event	0x1f0003	-
880	svchost.exe	0xe1a150f0	0x1b4	Key	0xf003f	
MACHINE\SOFTWARE\MICROSOFT\COM3						
880	svchost.exe	0x89967e28	0x1b8	Event	0x1f0003	-
880	svchost.exe	0xe1a901d0	0x1bc	Key	0x10 USER	
880	svchost.exe	0x89967df8	0x1c0	Event	0x1f0003	-
880	svchost.exe	0xe1a2b1e0	0x1c4	Key	0xf003f	
MACHINE\SOFTWARE\MICROSOFT\COM3						
880	svchost.exe	0x89968f80	0x1c8	Event	0x1f0003	-
880	svchost.exe	0xe18c3760	0x1cc	Key	0xf003f	
MACHINE\SOFTWARE\MICROSOFT\COM3						
880	svchost.exe	0x89968f18	0x1d0	Event	0x1f0003	-
880	svchost.exe	0xe1a8f458	0x1d4	Key	0xf003f	
MACHINE\SOFTWARE\CLASSES\CLSID						
880	svchost.exe	0x89969fb8	0x1d8	Event	0x1f0003	-
880	svchost.exe	0x899706e0	0x1dc	Event	0x1f0003	-
880	svchost.exe	0x896c4270	0x1e0	Thread	0x1f03ff	Tid 1744 Pid 880
880	svchost.exe	0xe1b1d660	0x1e4	Key	0xf003f	
MACHINE\SYSTEM\CONTROLSET001\SERVICES\TERMSERVICE\PARAMETERS						
880	svchost.exe	0xe1af1290	0x1e8	Port	0x1f0001	-
880	svchost.exe	0x8999a768	0x1ec	Event	0x1f0003	-
880	svchost.exe	0xe1b0a6b0	0x1f0	Key	0xf003f	
MACHINE\SYSTEM\CONTROLSET001\CONTROL\TERMINAL SERVER\LICENSING CORE						
880	svchost.exe	0x896a6f50	0x1f4	Event	0x1f0003	-
880	svchost.exe	0x8969ffa0	0x1f8	Event	0x1f0003	-

880	svchost.exe	0x89721c50	0x1fc	Mutant	0x1f0001	-
880	svchost.exe	0x896a8ed0	0x200	Event	0x1f0003	-
880	svchost.exe	0x896a8ea0	0x204	Event	0x1f0003	-
880	svchost.exe	0x8969bee0	0x208	Mutant	0x1f0001	-
880	svchost.exe	0x896a8e70	0x20c	Event	0x1f0003	-
880	svchost.exe	0x896a6db0	0x210	Mutant	0x1f0001	-
880	svchost.exe	0x899ea718	0x214	Event	0x1f0003	crypt32LogoffEvent
880	svchost.exe	0x896c1bc0	0x218	Mutant	0x1f0001	-
880	svchost.exe	0x896a8e40	0x21c	Event	0x1f0003	-
880	svchost.exe	0x896a8e00	0x220	Mutant	0x1f0001	-
880	svchost.exe	0x896a8dd0	0x224	Event	0x1f0003	-
880	svchost.exe	0x896a8da0	0x228	Event	0x1f0003	-
880	svchost.exe	0x89aed340	0x22c	Event	0x1f0003	TermSrvReadyEvent
880	svchost.exe	0x89994540	0x230	Semaphore	0x100003	-
880	svchost.exe	0x896a6f80	0x234	Semaphore	0x100003	-
880	svchost.exe	0x89957658	0x238	Mutant	0x1f0001	746bbf3569adEncrypt
880	svchost.exe	0xe14a68d8	0x23c	Port	0x1f0001	IcaApi
880	svchost.exe	0x896a2020	0x240	Thread	0x1f03ff	Tid 412 Pid 880
880	svchost.exe	0x8969ff70	0x244	Event	0x1f0003	-
880	svchost.exe	0x89695d30	0x248	Event	0x1f0003	-
880	svchost.exe	0xe1a9c388	0x24c	Key	0xf003f	

MACHINE\SOFTWARE\MICROSOFT\WINDOWS NT\CURRENTVERSION\WINLOGON

880	svchost.exe	0x8999ae20	0x250	Mutant	0x1f0001	-
880	svchost.exe	0xe1ad7298	0x254	Port	0x1f0001	-
880	svchost.exe	0x89693020	0x258	Event	0x1f0003	-
880	svchost.exe	0x896a5f18	0x25c	Event	0x1f0003	-
880	svchost.exe	0x899709d0	0x260	Event	0x1f0003	-
880	svchost.exe	0x896938c8	0x264	Event	0x1f0003	-
880	svchost.exe	0x896c0108	0x268	Event	0x1f0003	-
880	svchost.exe	0xe135f6a8	0x26c	Port	0x1f0001	SmSsWinStationApiPort
880	svchost.exe	0x899d6108	0x270	Event	0x1f0003	-
880	svchost.exe	0x8969fee0	0x274	File	0x12019f	\Device\Termdd
880	svchost.exe	0x899d60d8	0x278	Event	0x1f0003	-
880	svchost.exe	0xe1ad7668	0x27c	Key	0xf003f	

MACHINE\SYSTEM\CONTROLSET001\SERVICES\WINSOCK2\PARAMETERS\PROTOCOL_CATALOG9

880	svchost.exe	0x899760d0	0x280	Event	0x1f0003	-
880	svchost.exe	0xe1b0c960	0x284	Key	0xf003f	

MACHINE\SYSTEM\CONTROLSET001\SERVICES\WINSOCK2\PARAMETERS\NAMESPACE_CATALOG5

880	svchost.exe	0xe1b03df0	0x288	Port	0x1f0001	-
880	svchost.exe	0xe1b277c0	0x28c	Port	0x1f0001	-
880	svchost.exe	0x899a16c0	0x290	Mutant	0x1f0001	-
880	svchost.exe	0x89ab3478	0x294	File	0x12019f	\Device\Termdd
880	svchost.exe	0x89699d78	0x298	Event	0x1f0003	

WinMMConsoleAudioEvent

880	svchost.exe	0x89ab3978	0x29c	File	0x12019f	\Device\Termdd
880	svchost.exe	0x89a98da0	0x2a0	Process	0x1f0fff	csrss.exe Pid 592

880	svchost.exe	0x89a88da0	0x2a4	Process	0x1f0fff	winlogon.exe Pid 616
880	svchost.exe	0x89ae4f88	0x2a8	Event	0x1f0003	ReconEvent
880	svchost.exe	0x89699d38	0x2ac	Event	0x1f0003	TermSrv: machine GP event
880	svchost.exe	0x89991e20	0x2b0	Event	0x1f0003	-
880	svchost.exe	0x8970ab30	0x2b4	Thread	0x1f03ff	Tid 296 Pid 880
880	svchost.exe	0x896bcd18	0x2b8	File	0x12019f	
	\Device\NamedPipe\Ctx_WinStation_API_service					
880	svchost.exe	0x8997a248	0x2bc	File	0x12019f	
	\Device\NamedPipe\Ctx_WinStation_API_service					
880	svchost.exe	0x8970ab00	0x2c0	Event	0x1f0003	-
880	svchost.exe	0x896c4da8	0x2c4	Thread	0x1f03ff	Tid 300 Pid 880
880	svchost.exe	0x89996f10	0x2c8	Event	0x1f0003	-
880	svchost.exe	0x8996c7f0	0x2cc	Event	0x100000	userenv: Machine Group
	Policy has been applied					
880	svchost.exe	0x8968c9e8	0x2d0	Thread	0x1f03ff	Tid 316 Pid 880
880	svchost.exe	0xe1b1d358	0x2d4	Key	0x20019	
	MACHINE\SYSTEM\CONTROLSET001\CONTROL\TERMINAL SERVER					
880	svchost.exe	0xe1afd7a8	0x2d8	Key	0x20019	
	MACHINE\SOFTWARE\POLICIES					
880	svchost.exe	0x899845c8	0x2dc	Event	0x1f0003	-
880	svchost.exe	0x89984598	0x2e0	Event	0x1f0003	-
880	svchost.exe	0x89a7c678	0x2e4	Event	0x1f0003	-
880	svchost.exe	0xe1b11760	0x2e8	Port	0x1f0001	-
880	svchost.exe	0x8968c770	0x2ec	Thread	0x1f03ff	Tid 380 Pid 880
880	svchost.exe	0xe19917a0	0x2f0	Port	0x1f0001	-
880	svchost.exe	0xe1b116a8	0x2f4	Port	0x1f0001	-
880	svchost.exe	0xe1b1a020	0x2f8	Port	0x1f0001	-
880	svchost.exe	0xe1a15d48	0x2fc	Token	0xf01ff	-
880	svchost.exe	0x899dfca0	0x300	Event	0x100003	-
880	svchost.exe	0x899a24b0	0x304	File	0x12019f	\Device\Termdd
880	svchost.exe	0x899a3df8	0x308	Event	0x1f0003	-
880	svchost.exe	0x89695ab8	0x30c	Thread	0x1f03ff	Tid 236 Pid 880
880	svchost.exe	0xe1adf218	0x310	Port	0x1f0001	-
880	svchost.exe	0xe1aee2e0	0x314	Port	0x1f0001	-
880	svchost.exe	0xe1b30318	0x318	Port	0x1f0001	-
880	svchost.exe	0xe1ba5770	0x31c	Port	0x1f0001	-
880	svchost.exe	0x896c4240	0x320	Event	0x1f0003	-
880	svchost.exe	0x8968c020	0x324	Thread	0x1f03ff	Tid 1836 Pid 880
880	svchost.exe	0x89a71f58	0x32c	Event	0x100003	-
880	svchost.exe	0x896862b8	0x330	Mutant	0x1f0001	
	{3D5A1694-CC2C-4ee7-A3D5-A879A9E3A623}					
880	svchost.exe	0x89a0da50	0x334	Thread	0x1f03ff	Tid 1704 Pid 880
880	svchost.exe	0x89b9d840	0x338	Mutant	0x1f0001	-
880	svchost.exe	0x89a00f90	0x33c	File	0x12019f	
	\Device\{9DD6AFA1-8646-4720-836B-EDCB1085864A}					

880 svchost.exe 0x89af0cf0 0x340 File 0x12019f
IDevice\HarddiskVolume1\WINDOWS\system32\drivers\str.sys

880	svchost.exe	0xe1155570	0x344	Key	0xf003f	
MACHINE\SOFTWARE\CLASSES						
880	svchost.exe	0xe1139bb0	0x348	Key	0xf003f	
MACHINE\SOFTWARE\CLASSES						
880	svchost.exe	0x89986d98	0x34c	Event	0x1f0003	-
880	svchost.exe	0xe1bb4cf0	0x350	Key	0xf003f	
MACHINE\SOFTWARE\MICROSOFT\COM3						
880	svchost.exe	0x89986dc8	0x354	Event	0x1f0003	-
880	svchost.exe	0xe1b0e968	0x358	Key	0x10	USER
880	svchost.exe	0x89a7b930	0x35c	Event	0x1f0003	-
880	svchost.exe	0xe118b468	0x360	Key	0xf003f	
MACHINE\SOFTWARE\CLASSES						
880	svchost.exe	0x89ba1628	0x364	Event	0x1f0003	-
880	svchost.exe	0xe16b73c8	0x368	Key	0x10	USER
880	svchost.exe	0x899e1f18	0x36c	Event	0x1f0003	-
880	svchost.exe	0xe2086a88	0x370	Key	0xf003f	
MACHINE\SOFTWARE\MICROSOFT\COM3						
880	svchost.exe	0x89a492b0	0x374	Event	0x1f0003	-
880	svchost.exe	0xe2086af0	0x378	Key	0xf003f	
MACHINE\SOFTWARE\MICROSOFT\COM3						
880	svchost.exe	0x89b9d880	0x37c	Event	0x1f0003	-
880	svchost.exe	0xe1b4a3c8	0x380	Key	0xf003f	
MACHINE\SOFTWARE\CLASSES\CLSID						
880	svchost.exe	0x8969fc30	0x384	Event	0x1f0003	-
880	svchost.exe	0xe1b4a360	0x388	Key	0xf003f	
MACHINE\SOFTWARE\CLASSES						
880	svchost.exe	0x89c0a9e8	0x38c	Event	0x1f0003	-
880	svchost.exe	0xe16b7360	0x390	Key	0xf003f	
MACHINE\SOFTWARE\MICROSOFT\COM3						
880	svchost.exe	0x89bd3db0	0x394	Event	0x1f0003	-
880	svchost.exe	0xe19b8140	0x398	Key	0x10	USER
880	svchost.exe	0x89b961e0	0x39c	Event	0x1f0003	-
880	svchost.exe	0xe16a0fb8	0x3a0	Key	0xf003f	
MACHINE\SOFTWARE\MICROSOFT\COM3						
880	svchost.exe	0x89bdd318	0x3a4	Event	0x1f0003	-
880	svchost.exe	0xe16a0f50	0x3a8	Key	0xf003f	
MACHINE\SOFTWARE\MICROSOFT\COM3						
880	svchost.exe	0x89aac2c0	0x3ac	Event	0x1f0003	-
880	svchost.exe	0xe1bb4678	0x3b0	Key	0xf003f	
MACHINE\SOFTWARE\CLASSES\CLSID						
880	svchost.exe	0x89bd23f8	0x3b4	Event	0x1f0003	-
880	svchost.exe	0x89a0da50	0x3b8	Thread	0x4a	Tid 1704 Pid 880
880	svchost.exe	0xe1a0f360	0x3bc	Key	0x20019	
MACHINE\SOFTWARE\CLASSES						
880	svchost.exe	0x89a8e248	0x3c0	Event	0x1f0003	-

```

880  svchost.exe  0x899f71b0  0x3c4  Semaphore  0x1f0003  -
880  svchost.exe  0x89bc46f0  0x3c8  Semaphore  0x1f0003  -
880  svchost.exe  0x899e7f18  0x3cc  Semaphore  0x1f0003  -
880  svchost.exe  0x899e7ee8  0x3d0  Event  0x1f0003  -
880  svchost.exe  0xe1a8fb10  0x3d4  Key  0x20f003f  USER\DEFAULT
880  svchost.exe  0x89993f90  0x3d8  File  0x100020
\Device\HarddiskVolume1\WINDOWS\WinSxS\x86_Microsoft.Windows.Common-Controls_6
595b64144ccf1df_6.0.2600.5512_x-ww_35d4ce83
880  svchost.exe  0xe1a0f3c8  0x3dc  Key  0x2001f
USER\DEFAULT\SOFTWARE\MICROSOFT\WINDOWS\CURRENTVERSION\INTERNET
SETTINGS
880  svchost.exe  0x89970590  0x3e0  Mutant  0x100000  _!MSFTHISTORY!_
880  svchost.exe  0x89958b78  0x3e4  File  0x12019f
\Device\HarddiskVolume1\WINDOWS\system32\config\systemprofile\Local
Settings\Temporary Internet Files\Content.IE5\index.dat
880  svchost.exe  0x897301d8  0x3e8  Mutant  0x1f0001
c:\windows\system32\config\systemprofile\local settings\temporary internet files\content.ie5\
880  svchost.exe  0xe1a29fd8  0x3ec  Section  0xf0007
C:_WINDOWS_system32_config_systemprofile_Local Settings_Temporary Internet
Files_Content.IE5_index.dat_32768
880  svchost.exe  0x89730228  0x3f0  Mutant  0x1f0001
c:\windows\system32\config\systemprofile\cookies\
880  svchost.exe  0x89731ee8  0x3f4  Semaphore  0x1f0003
shell.{210A4BA0-3AEA-1069-A2D9-08002B30309D}
880  svchost.exe  0x899fe2e0  0x3f8  File  0x12019f
\Device\HarddiskVolume1\WINDOWS\system32\config\systemprofile\Cookies\index.dat
880  svchost.exe  0x89a3a150  0x3fc  Mutant  0x1f0001
c:\windows\system32\config\systemprofile\local settings\history\history.ie5\
880  svchost.exe  0x89a492e8  0x400  File  0x12019f
\Device\HarddiskVolume1\WINDOWS\system32\config\systemprofile\Local
Settings\History\History.IE5\index.dat
880  svchost.exe  0xe2037cf8  0x404  Section  0xf0007
C:_WINDOWS_system32_config_systemprofile_Local
Settings_History_History.IE5_index.dat_32768
880  svchost.exe  0x899987a0  0x408  Mutant  0x100000  WininetStartupMutex
880  svchost.exe  0x8995b1e8  0x40c  Event  0x1f0003  -
880  svchost.exe  0xe1219650  0x410  Section  0xf0007
C:_WINDOWS_system32_config_systemprofile_Cookies_index.dat_16384
880  svchost.exe  0x89964648  0x414  Mutant  0x1f0001  -
880  svchost.exe  0x8970b208  0x418  Mutant  0x1f0001  -
880  svchost.exe  0x8996e0e0  0x41c  Mutant  0x100000
WininetProxyRegistryMutex
880  svchost.exe  0x896811d8  0x424  File  0x100020
\Device\HarddiskVolume1\WINDOWS\WinSxS\x86_Microsoft.Windows.Common-Controls_6
595b64144ccf1df_6.0.2600.5512_x-ww_35d4ce83
880  svchost.exe  0x89a4d980  0x428  Mutant  0x1f0001  -
880  svchost.exe  0x89965fe0  0x42c  Mutant  0x100000  RasPbFile
880  svchost.exe  0x8997fe90  0x430  Event  0x1f0003  -

```

```

880  svchost.exe  0x89a4d950  0x434  Event  0x1f0003  -
880  svchost.exe  0x89996e60  0x438  Event  0x1f0003  -
880  svchost.exe  0x89a7f110  0x43c  Semaphore  0x100003  -
880  svchost.exe  0x89698d08  0x440  Semaphore  0x100003  -
880  svchost.exe  0x89996e30  0x444  Event  0x1f0003  -
880  svchost.exe  0x89996e00  0x448  Event  0x1f0003  -
880  svchost.exe  0x89b961a8  0x44c  Semaphore  0x100003  -
880  svchost.exe  0x89b9d280  0x450  Semaphore  0x100003  -
880  svchost.exe  0xe12195c8  0x454  Key  0x20019
MACHINE\SOFTWARE\MICROSOFT\TRACING\RASAPI32
880  svchost.exe  0x89ab0718  0x458  Event  0x1f0003  -
880  svchost.exe  0x899e5888  0x45c  Event  0x1f0003  -
880  svchost.exe  0x89a0da50  0x460  Thread  0x1f03ff  Tid 1704 Pid 880
880  svchost.exe  0x89675cd0  0x464  Event  0x1f0003  -
880  svchost.exe  0xe17147b0  0x468  Section 0x4  SENS Information Cache
880  svchost.exe  0xe14e3728  0x46c  Port  0x1f0001  -
880  svchost.exe  0xe15f0d68  0x470  Key  0x20f003f  USER
880  svchost.exe  0xe169e898  0x474  Key  0xf003f
MACHINE\SYSTEM\CONTROLSET001\HARDWARE PROFILES\0001
880  svchost.exe  0x89a76f60  0x478  Mutant  0x1f0001
ZonesLockedCacheCounterMutex
880  svchost.exe  0x896836b0  0x47c  Mutant  0x1f0001
ZonesCacheCounterMutex
880  svchost.exe  0x89683700  0x480  Mutant  0x1f0001  ZonesCounterMutex
880  svchost.exe  0xe19b6a48  0x484  Key  0x20019
USER\DEFAULT\SOFTWARE\MICROSOFT\WINDOWS\CURRENTVERSION\INTERNET
SETTINGS\ZONEMAP
880  svchost.exe  0x89bbc028  0x488  File  0x100020
\Device\HarddiskVolume1\WINDOWS\WinSxS\x86_Microsoft.Windows.Common-Controls_6
595b64144ccf1df_6.0.2600.5512_x-ww_35d4ce83
880  svchost.exe  0xe11984b8  0x48c  Key  0x20019
USER\DEFAULT\SOFTWARE\MICROSOFT\WINDOWS\CURRENTVERSION\INTERNET
SETTINGS\ZONEMAP
880  svchost.exe  0xe11f4890  0x490  Section 0xf0007 UrlZonesSM_SYSTEM
880  svchost.exe  0x89a0da50  0x494  Thread  0x1f03ff  Tid 1704 Pid 880
880  svchost.exe  0x899a1160  0x498  Event  0x1f0003  -
880  svchost.exe  0x8999c6d8  0x49c  Event  0x1f0003  -
880  svchost.exe  0xe1bc5588  0x4a0  Port  0x1f0001  -
880  svchost.exe  0x89999980  0x4a8  File  0x1200a0
\Device\NetBT_Tcpip_{B35F0A5F-EBC3-4B5D-800D-7C1B64B30F14}
FLARE-VM 04/11/2026 12:14:36

```

PS C:\Tools\volatility3-win-exes-2.27.0 > .\vol.exe -f

C:\Users\user_lab\Desktop\temp_extract_dir\CYBERDEF-567078-20230213-171333.raw

windows.dlllist --pid 880

Volatility 3 Framework 2.27.0

Progress: 100.00 PDB scanning finished

PID	Process	Base	Size	Name	Path	LoadCount	LoadTime	File output
880	svchost.exe	0x1000000	0x6000	svchost.exe	C:\WINDOWS\system32\svchost.exe	-1	N/A	Disabled
880	svchost.exe	0x7c900000	0xaf000	ntdll.dll	C:\WINDOWS\system32\ntdll.dll	-1	N/A	Disabled
880	svchost.exe	0x7c800000	0xf6000	kernel32.dll	C:\WINDOWS\system32\kernel32.dll	-1	N/A	Disabled
880	svchost.exe	0x77dd0000	0x9b000	ADVAPI32.dll	C:\WINDOWS\system32\ADVAPI32.dll	-1	N/A	Disabled
880	svchost.exe	0x77e70000	0x92000	RPCRT4.dll	C:\WINDOWS\system32\RPCRT4.dll	-1	N/A	Disabled
880	svchost.exe	0x77fe0000	0x11000	Secur32.dll	C:\WINDOWS\system32\Secur32.dll	-1	N/A	Disabled
880	svchost.exe	0x5cb70000	0x26000	ShimEng.dll	C:\WINDOWS\system32\ShimEng.dll	1	N/A	Disabled
880	svchost.exe	0x6f880000	0x1ca000	AcGenral.DLL	C:\WINDOWS\AppPatch\AcGenral.DLL	1	N/A	Disabled
880	svchost.exe	0x7e410000	0x91000	USER32.dll	C:\WINDOWS\system32\USER32.dll	844	N/A	Disabled
880	svchost.exe	0x77f10000	0x49000	GDI32.dll	C:\WINDOWS\system32\GDI32.dll	502	N/A	Disabled
880	svchost.exe	0x76b40000	0x2d000	WINMM.dll	C:\WINDOWS\system32\WINMM.dll	4	N/A	Disabled
880	svchost.exe	0x774e0000	0x13d000	ole32.dll	C:\WINDOWS\system32\ole32.dll	126	N/A	Disabled
880	svchost.exe	0x77c10000	0x58000	msvcrt.dll	C:\WINDOWS\system32\msvcrt.dll	750	N/A	Disabled
880	svchost.exe	0x77120000	0x8b000	OLEAUT32.dll	C:\WINDOWS\system32\OLEAUT32.dll	116	N/A	Disabled
880	svchost.exe	0x77be0000	0x15000	MSACM32.dll	C:\WINDOWS\system32\MSACM32.dll	1	N/A	Disabled
880	svchost.exe	0x77c00000	0x8000	VERSION.dll	C:\WINDOWS\system32\VERSION.dll	7	N/A	Disabled
880	svchost.exe	0x7c9c0000	0x817000	SHELL32.dll	C:\WINDOWS\system32\SHELL32.dll	3	N/A	Disabled
880	svchost.exe	0x77f60000	0x76000	SHLWAPI.dll	C:\WINDOWS\system32\SHLWAPI.dll	118	N/A	Disabled
880	svchost.exe	0x769c0000	0xb4000	USERENV.dll	C:\WINDOWS\system32\USERENV.dll	4	N/A	Disabled
880	svchost.exe	0x5ad70000	0x38000	UxTheme.dll	C:\WINDOWS\system32\UxTheme.dll	1	N/A	Disabled

880 svchost.exe 0x773d0000 0x103000 comctl32.dll
 C:\WINDOWS\WinSxS\x86_Microsoft.Windows.Common-Controls_6595b64144ccf1df_6.0.2
 600.5512_x-ww_35d4ce83\comctl32.dll 4 N/A Disabled
 880 svchost.exe 0x5d090000 0x9a000 comctl32.dll
 C:\WINDOWS\system32\comctl32.dll 1 N/A Disabled
 880 svchost.exe 0x77690000 0x21000 NTMARTA.DLL
 C:\WINDOWS\system32\NTMARTA.DLL 1 N/A Disabled
 880 svchost.exe 0x71bf0000 0x13000 SAMLIB.dll
 C:\WINDOWS\system32\SAMLIB.dll 1 N/A Disabled
 880 svchost.exe 0x76f60000 0x2c000 WLDAP32.dll
 C:\WINDOWS\system32\WLDAP32.dll 2 N/A Disabled
 880 svchost.exe 0x76a80000 0x64000 rpcss.dll c:\windows\system32\rpcss.dll
 1 N/A Disabled
 880 svchost.exe 0x71ab0000 0x17000 WS2_32.dll
 c:\windows\system32\WS2_32.dll 35 N/A Disabled
 880 svchost.exe 0x71aa0000 0x8000 WS2HELP.dll
 c:\windows\system32\WS2HELP.dll 36 N/A Disabled
 880 svchost.exe 0x670000 0x2c5000 xpsp2res.dll
 C:\WINDOWS\system32\xpsp2res.dll 1 N/A Disabled
 880 svchost.exe 0x76fd0000 0x7f000 CLBCATQ.DLL
 C:\WINDOWS\system32\CLBCATQ.DLL 4 N/A Disabled
 880 svchost.exe 0x77050000 0xc5000 COMRes.dll
 C:\WINDOWS\system32\COMRes.dll 4 N/A Disabled
 880 svchost.exe 0x760f0000 0x53000 termsrv.dll c:\windows\system32\termsrv.dll
 1 N/A Disabled
 880 svchost.exe 0x74f70000 0x6000 ICAAPI.dll c:\windows\system32\ICAAPI.dll
 1 N/A Disabled
 880 svchost.exe 0x77920000 0xf3000 SETUPAPI.dll
 c:\windows\system32\SETUPAPI.dll 1 N/A Disabled
 880 svchost.exe 0x76c30000 0x2e000 WINTRUST.dll
 C:\WINDOWS\system32\WINTRUST.dll 1 N/A Disabled
 880 svchost.exe 0x77a80000 0x95000 CRYPT32.dll
 C:\WINDOWS\system32\CRYPT32.dll 108 N/A Disabled
 880 svchost.exe 0x77b20000 0x12000 MSASN1.dll
 C:\WINDOWS\system32\MSASN1.dll 107 N/A Disabled
 880 svchost.exe 0x76c90000 0x28000 IMAGEHLP.dll
 C:\WINDOWS\system32\IMAGEHLP.dll 1 N/A Disabled
 880 svchost.exe 0x776c0000 0x12000 AUTHZ.dll
 c:\windows\system32\AUTHZ.dll 1 N/A Disabled
 880 svchost.exe 0x75110000 0x1f000 mstlsapi.dll
 c:\windows\system32\mstlsapi.dll 1 N/A Disabled
 880 svchost.exe 0x77cc0000 0x32000 ACTIVEDS.dll
 c:\windows\system32\ACTIVEDS.dll 1 N/A Disabled
 880 svchost.exe 0x76e10000 0x25000 adslidpc.dll
 c:\windows\system32\adslidpc.dll 1 N/A Disabled
 880 svchost.exe 0x5b860000 0x55000 NETAPI32.dll
 C:\WINDOWS\system32\NETAPI32.dll 5 N/A Disabled

```

880  svchost.exe  0x76b20000  0x11000 ATL.DLL c:\windows\system32\ATL.DLL  1
N/A  Disabled
880  svchost.exe  0x76bc0000  0xf000 REGAPI.dll
C:\WINDOWS\system32\REGAPI.dll 1  N/A  Disabled
880  svchost.exe  0x68000000  0x36000 rsaenh.dll
C:\WINDOWS\system32\rsaenh.dll 1  N/A  Disabled
880  svchost.exe  0x74980000  0x113000 msxml3.dll
C:\WINDOWS\system32\msxml3.dll 1  N/A  Disabled
880  svchost.exe  0x771b0000  0xaa000 wininet.dll
C:\WINDOWS\system32\wininet.dll 105  N/A  Disabled
880  svchost.exe  0x71ad0000  0x9000 wsock32.dll
C:\WINDOWS\system32\wsock32.dll 1  N/A  Disabled
880  svchost.exe  0x76ee0000  0x3c000 RASAPI32.DLL
C:\WINDOWS\system32\RASAPI32.DLL 2  N/A  Disabled
880  svchost.exe  0x76e90000  0x12000 rasman.dll
C:\WINDOWS\system32\rasman.dll 2  N/A  Disabled
880  svchost.exe  0x76eb0000  0x2f000 TAPI32.dll
C:\WINDOWS\system32\TAPI32.dll 2  N/A  Disabled
880  svchost.exe  0x76e80000  0xe000 rtutils.dll C:\WINDOWS\system32\rtutils.dll
3  N/A  Disabled
880  svchost.exe  0x722b0000  0x5000 sensapi.dll
C:\WINDOWS\system32\sensapi.dll 1  N/A  Disabled
880  svchost.exe  0x7e1e0000  0xa2000 urlmon.dll
C:\WINDOWS\system32\urlmon.dll 1  N/A  Disabled
880  svchost.exe  0x71a50000  0x3f000 mswsock.dll
C:\WINDOWS\System32\mswsock.dll 1  N/A  Disabled
880  svchost.exe  0x76f20000  0x27000 DNSAPI.dll
C:\WINDOWS\system32\DNSAPI.dll 1  N/A  Disabled
880  svchost.exe  0x76fc0000  0x6000 rasadhlp.dll
C:\WINDOWS\system32\rasadhlp.dll 1  N/A  Disabled
FLARE-VM 04/11/2026 12:54:26

```

Q1 ✓ Solved : 3030

Which volatility profile would be best for this machine?

WinXPSP2x86

💡 Hints

▶ Submit

Q2 ✓ Solved : 2972

How many processes were running when the image was acquired?

**
19

💡 Hints

▶ Submit

Q3 ✓ Solved : 3077

What is the process ID of `cmd.exe` ?

1960

💡 Hints

▶ Submit

Q4 ✓ Solved : 3063

What is the name of the most suspicious process?

rootkit.exe

💡 Hints

▶ Submit

Q5 ✓ Solved : 3000

Which process shows the highest likelihood of code injection?

svchost.exe

💡 Hints

▶ Submit

Q6 ✓ Solved : 2776

There is an odd file referenced in the recent process. Provide the full path of that file.

*.******************
C:\WINDOWS\system32\drivers\str.sys

💡 Hints

▶ Submit

Q7 ✓ Solved : 2746

What is the name of the injected DLL file loaded from the recent process?

msxml3r.dll

💡 Hints

▶ Submit

Q8 ✓ Solved : 2728

What is the base address of the injected DLL?

0x980000

💡 Hints

▶ Submit